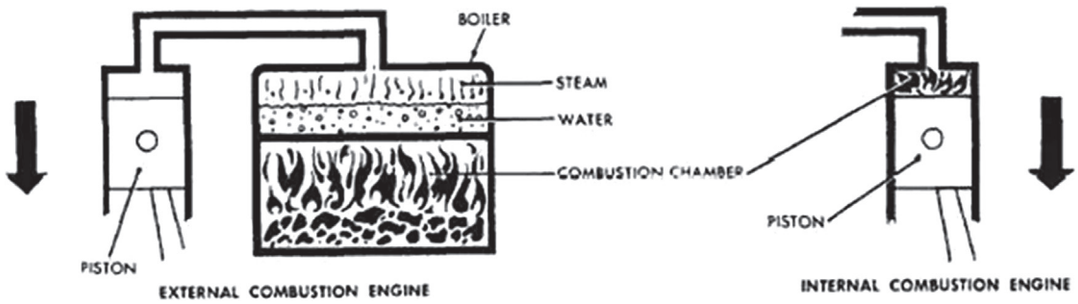


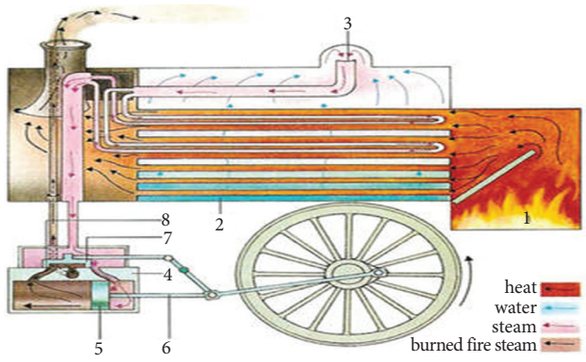
Figure 4.2

4.2.1 External Combustion Engine

An external combustion engine is a heat engine where a working fluid, is heated by combustion in an external source like Boiler. The fluid then, by expanding and acting on the mechanism of the engine, produces motion and usable work.

In external combustion engines, the combustion process takes place outside the mechanical engine system. External Combustion Engines are used in the following.

Ancient Marine Engine



Ancient Road Roller Engine

Steam Locomotive

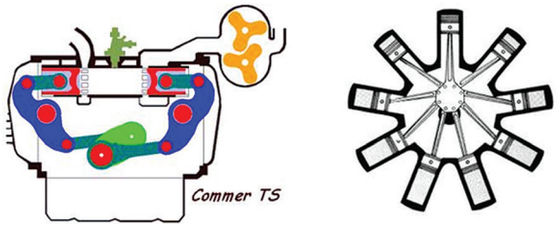


Figure 4.2.1

4.2.2 INTERNAL COMBUSTION ENGINE

An internal combustion engine (ICE) is a heat engine where the combustion of a fuel and air occurs inside the engine combustion chamber that is an integral part of an engine.

4.2.3 Classification of Internal Combustion Engine

Internal Combustion Engines are classified in many different ways as follows

1. According to the Cycle of Operation
   a) Otto Cycle
   b) Diesel Cycle
2. According to the No. of Stroke
   a) Two stroke engine
   b) Four stroke engine

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